CLAIMS

What is claimed is:

1. A data transmission apparatus for spatially-encoded optical data, comprising:
   - a flexible cable containing a coherent fiber optic bundle, said bundle comprising a plurality of optical fibers; and
   - wherein the relative spatial position of each optical fiber in the bundle is maintained from a first end of the cable to a second end of the cable, such that an image projected onto the first end is replicated at the second end.

2. The apparatus of claim 1, wherein the cable is configured to optically couple a two-dimensional array of photon emitters at the first end with a two-dimensional array of photon sensors at the second end.

3. The apparatus of claim 1, wherein the apparatus is configured to transmit data at a rate exceeding 1 terabit per second.

4. A data transmission system, comprising:
   - a transmitter module containing a two-dimensional array of photon emitters;
   - a receiver module containing a two-dimensional array of photon sensors; and
   - a flexible, coherent fiber optic bundle physically connecting the transmitter module and the receiver module, configured to preserve the spatial pattern of photons from the emitter array to the sensor array.

5. A method for transmitting data, comprising the steps of:
   - generating a two-dimensional frame of light using an array of optical emitters, wherein data is encoded in the spatial characteristics of the frame;
   - projecting said frame onto a first face of a coherent fiber optic bundle;

- transmitting the frame through the bundle, whereby the spatial arrangement of the frame is preserved; and
- receiving the frame at a second face of the bundle with an array of optical sensors.

6. A rigid data interconnect apparatus, comprising:
   - a sealed, light-proof housing having a first end and a second end;
   - a first port at the first end, configured to receive light from an external two-dimensional array of photon emitters;
   - at least one collimating lens positioned near the first end, configured to convert diverging light from the emitters into a parallel beam;
   - at least one focusing lens positioned near the second end, configured to converge the parallel beam; and
   - a second port at the second end, configured to project the converged beam onto an external two-dimensional array of photon sensors.

7. The apparatus of claim 6, wherein the interconnect establishes a free-space optical path between the first port and the second port.

8. The apparatus of claim 6, wherein the interconnect is configured for short-range communication between electronic components within a server rack.

9. A data transmission system, comprising:
   - a transmitter module containing an array of emitters;
   - a receiver module containing an array of sensors; and
   - a rigid, sealed interconnect optically coupling the transmitter and receiver, said interconnect containing a system of lenses to project frames of light from the emitters onto the sensors.

10. The system of claim 9, wherein the system of lenses comprises at least one collimating lens and at least one focusing lens.